a) Equipment Inventory

$$serialNo \to \{modelId, labId\} \quad \text{and} \quad modelId \to manufacturer$$

$$serialNo \to modelId \quad \text{and} \quad modelId \to manufacturer$$

$$serialNo \to manufacturer \quad \text{(par la transitivité)}$$

$$serialNo \to \{modelId, labId, manufacturer\}$$

- La relation ne satisfait pas BCNF car $modelId \to manufacturer$ et $modelId$ n'est pas une superclé.
- N'est pas 3NF : $manufacturer \notin$ candidate key $= \{serialNo\}$.

Strongest normal form: 1NF

Décomposition : $modelId \to manufacturer$

$$(modelId)^+ = \{manufacturer, modelId\}$$

$$R_1 = (\underbrace{\underline{modelId}}_{\text{(Primary Key)}}, manufacturer) = X^+ \qquad \textbf{BCNF}$$

$$R_2 = (modelId, \underbrace{\underline{serialNo}}_{\text{(Primary Key)}}, labId) \qquad \textbf{BCNF}$$

On a :

$$F_{R_1} = \{\, modelId \to manufacturer \,\}$$
$$F_{R_2} = \{\, serialNo \to \{modelId, labId\} \,\}$$

D'où : **FD are preserved.**

b) Project Participation

Candidate key is $\{pid, projectCode\}$.

Comme

$$\{pid, projectCode\} \to \{pid, projectCode, role, piPid\}$$

la relation ne satisfait pas BCNF vu que $projectCode \to piPid$ et $projectCode$ is not a superkey ($projectCode \not\to pid$).

N'est pas 3NF : $piPid \notin$ candidate key $= \{pid, projectCode\}$.

D'où : **Strongest normal form:** 1NF

$$(projectCode)^+ = \{projectCode, piPid\}$$

$$R_1 = (\underbrace{\underline{projectCode}}_{\text{(Primary Key)}}, \ piPid) \qquad \textbf{BCNF}$$

$$R_2 = (\underbrace{\underline{projectCode}, \underline{pid}}_{\text{(Primary Key)}}, \ role) \qquad \textbf{BCNF}$$

$$F_{R_1} = \{\, projectCode \rightarrow piPid \,\}$$
$$F_{R_2} = \{\, (pid, projectCode) \rightarrow role \,\}$$

D'où : **FD are preserved.**

c) Equipment Certification Requirement

Candidate key is $\{modelId, certCode\}$.

Because $certCode \rightarrow \{certTitle, issuingAuthority\}$, the relation is not BCNF vu que $certCode$ n'est pas une superclé ; en effet $certCode \not\rightarrow modelId$.

Strongest normal form: 1NF, 3NF vu que $certCode \in$ candidate key.

$$R_1 = (certCode)^+ = \{certCode, certTitle, issuingAuthority\} \qquad \textbf{BCNF}$$
$$R_2 = (certCode) \cup (R - X^+) = (certCode) \cup (R - R_1) = \{certCode, modelId\} \qquad \textbf{BCNF}$$

$$F_{R_1} = \{\, certCode \rightarrow \{certTitle, issuingAuthority\} \,\}$$
$$F_{R_2} = \{\, (certCode, modelId) \rightarrow certCode \text{ et } modelId \,\}$$

$$(F_{R_1} \cup F_{R_2})^+ = \Big\{ (certCode, modelId) \rightarrow \{certTitle, issuingAuthority, certCode, modelId\},$$
$$certCode \rightarrow \{certTitle, issuingAuthority\} \Big\}$$

Donc $(F_{R_1} \cup F_{R_2})^+ = F^+$.

D'où : **FD are preserved.**

d) Supplier Offer

Candidate key is $\{suppId, consId, labId\}$.

Because $(suppId, consId, labId) \rightarrow suppId \rightarrow supplierName$, the relation is not BCNF vu que $suppId$ is not a superkey ; en effet $suppId \not\rightarrow consId$ or $labId$.

Strongest normal form: 1NF, 3NF comme $suppId \in$ candidate key.

$$R_1 = (suppId)^+ = \{ \underbrace{suppId}_{\text{(Primary Key)}}, supplierName\} \quad \textbf{BCNF}$$

$$R_2 = \{\underbrace{suppId, consId, labId}_{\text{(Primary Key)}}, unitPrice\} \quad \textbf{BCNF}$$

De même :

$$F_{R_1} = \{ suppId \rightarrow \{suppId, supplierName\} \}$$
$$F_{R_2} = \{ \{suppId, consId, labId\} \rightarrow unitPrice \}$$

Donc $(F_{R_1} \cup F_{R_2})^+ = F^+$.

D'où : **FD are preserved.**

e) Reservation and Project Information

Candidate key is $\{resId\}$.

Because $resId \rightarrow projectCode \rightarrow projectTitle$, et encore $resId \rightarrow \{projectCode, madeByPid, status\}$, the relation is not BCNF vu que $projectCode$ is not a superkey ; en effet $projectCode \not\rightarrow resId$.

Strongest normal form: 1NF

Is not 3NF comme $projectCode \notin$ candidate key.

$$R_1 = (projectCode)^+ = \{ \underbrace{projectCode}_{\text{(Primary Key)}}, projectTitle\} \quad \textbf{BCNF}$$

$$R_2 = \{projectCode, \underbrace{resId}_{\text{(Primary Key)}}, madeByPid, status\} \quad \textbf{BCNF}$$

De même :

$$F_{R_1} = \{ projectCode \rightarrow projectTitle \}$$
$$F_{R_2} = \{ resId \rightarrow \{resId, projectCode, madeByPid, status\} \}$$

Donc $(F_{R_1} \cup F_{R_2})^+ = F^+$.

D'où : **FD are preserved.**